SVD Weaving Draft Explorer — Complete Tutorial

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Introduction

This tool takes any black-and-white bitmap image and factors it into three matrices — threading, tie-up, and treadling — producing a standard weaving draft with a drawdown. Five factorization methods are available. You can edit the draft by hand after computation, flip the face, assign colours to warp and weft, and export to WIF for Fiberworks, PixeLoom, or any WIF-compatible software.

The page is designed for students exploring the connection between matrix factorization and weaving structure. The central idea: a binary image is a matrix, and factorizing it into three pieces corresponds exactly to the three components of a weaving draft.

What you need: A black-and-white or greyscale BMP file. Images from 14×14 to 60×60 pixels work well. Very large images are slower to compute, especially with alternating optimisation. Most image editors (GIMP, Photoshop, Paint) can save to BMP.

This tool takes any black-and-white bitmap image, factors it into three matrices using one of five methods, and displays the result as a weaving draft — threading, tie-up, treadling, and drawdown. You can edit the draft by hand after computation, flip the face, assign colours to warp and weft, and export to WIF for Fiberworks, PixeLoom, or any WIF-compatible software.

This tutorial covers the advanced features. The basic tutorial covers image upload, scaling, and core SVD factorization; this document focuses on the additional factorization methods, assignment options, manual editing, colour schemes, and exporting.

What you need: A black-and-white or greyscale BMP file, 14×14 to 60×60 pixels. Very large images are slower, especially with alternating optimisation.

Step 1: Upload and scale the image

Click the upload area or drag a BMP file onto it. The page accepts 1-bit (pure black and white), 4-bit, 8-bit greyscale, 24-bit colour, and 32-bit colour BMP files. Colour images are converted automatically: pixels below 50% brightness become 1 (dark, warp up) and pixels above become 0 (light, weft up). Once loaded, two dimensions appear:

- **X** — width in pixels, which becomes the number of warp threads
- **Y** — height in pixels, which becomes the number of picks (weft rows)

The **Scale** field (default 100%) resamples the image using nearest-neighbour interpolation. Enter any integer from 1 to 100 and click Apply. The original upload is always kept in memory, so you can move the scale freely without losing quality. Scale the image before editing pixels — changing the scale after editing discards your edits.

Tip: Start small. A 14×14 image at Z=14 computes in under a second with any method. Scaling to 50% before factoring is a quick way to explore the draft structure before committing to the full resolution.

Click the upload area or drag a BMP file onto it. The page accepts 1-bit, 8-bit, 24-bit, and 32-bit BMP files. Greyscale and colour images are converted automatically: pixels below 50% brightness become 1 (dark, warp up) and pixels above become 0 (light, weft up). Once loaded, the dimensions X (warps) and Y (picks) appear above the matrix view.

The **Scale** field (default 100%) resamples the image using nearest-neighbour interpolation. The original is always kept in memory, so you can move the scale freely. Scale the image before editing pixels — changing the scale after editing discards your edits.

Tip: Start small. A 14×14 image at Z=14 computes in under a second with any method. Scaling to 50% before factoring is a good way to explore the draft structure before committing to the full resolution.

Step 2: Edit the matrix pixel by pixel

The matrix view shows the image as a grid of black (1) and white (0) squares. It is fully editable:

- **Click** any cell to toggle it between 0 and 1.
- **Click and drag** to paint a stroke. The paint value is the opposite of the first cell you click, so dragging across white cells turns them black and vice versa, without accidentally reversing cells you have already painted.
- **Touch support** is included — the same gestures work on a phone or tablet.

Edits apply to the scaled working matrix only. The original upload is preserved; rescaling resets your edits. Edit after scaling, not before.

This editor is useful for fixing small details in an uploaded image, or for drawing simple test patterns from scratch — for example, a checkerboard or diagonal stripe — to study how each factorization method handles structured binary images.

Step 3: Set Z and choose a factorization method

The **Method** dropdown selects how the image matrix is decomposed into threading, tie-up, and treadling. All methods take the same Z (number of shafts/treadles) as input and report reconstruction error so you can compare results directly.

Method	What it does	Best for
SVD	Singular value decomposition — mathematically optimal continuous approximation, then binarised	General use; full-rank reconstruction when Z = min(X, Y)
NMF	Non-negative matrix factorization — finds additive, parts-based factors	Images with distinct positive regions; avoids cancellation
BMF	Boolean matrix factorization — alternating optimisation directly on binary matrices	Maximally faithful binary reconstruction; slowest per restart

Method	What it does	Best for
K-means	Clusters warp columns into Z shaft groups and pick rows into Z treadle groups	Clean block-structured drafts; fast
Alternating optimisation	Starts from SVD, K-means, or random, then iterates to minimise error	Lowest error at small Z; best overall when restarts > 1

Each method except SVD has **Random restarts** — the computation runs N times from different starting points and keeps the result with the lowest reconstruction error. More restarts take longer but consistently find better solutions.

Max iterations controls how long each restart runs. The default values are conservative; increase them for better results at the cost of speed.

Step 4: Assignment options

Two checkboxes below the method selector apply to all methods and control how continuous factorization values are converted into the binary draft.

Allow multiple treadles per pick — when checked, the tool sweeps K from 1 to Z, evaluating reconstruction error at every level and keeping the K that gives the lowest error. This can never make results worse than single-treadle: K=1 is always a candidate and wins if no higher K improves on it. The average treadles per pick is shown in the status bar after computation. This option is unavailable for the Hungarian algorithm (see below) because that algorithm enforces a 1-to-1 pick-to-treadle assignment.

Hungarian algorithm (SVD only) — the standard SVD assignment uses argmax: each warp is assigned to the shaft with the largest absolute value in its singular vector, independently of all other warps. This is fast but greedy — multiple warps can share a shaft. The Hungarian algorithm finds the globally optimal 1-to-1 assignment, minimising total alignment loss. When Z = min(X, Y), it guarantees every warp is on a distinct shaft and every pick presses a distinct treadle, which allows perfect reconstruction.

- When both options are checked, a warning appears. Hungarian is still evaluated as a K=1 candidate in the multi-treadle sweep and wins on ties — so it is kept if it matches or beats the best non-Hungarian result.
- Hungarian is not used for K > 1 because 1-to-1 assignment prevents the treadle sharing that multi-treadle mode requires.

Step 5: Reading and editing the draft

The draft appears in the conventional weaving layout: threading top-left, tie-up top-right, treadling below the tie-up, drawdown bottom-left. The treadling and tie-up share the right column so treadle columns align — treadle 1 in the tie-up sits directly above treadle 1 in the treadling.

All three input matrices are editable after computation:

Threading (blue, rows = shafts, cols = warps) — click any cell to move that warp to a different shaft. The previous shaft assignment clears automatically. Exactly one shaft per warp is enforced at all times.

Tie-up (green, Z × Z) — click to toggle any cell between 0 and 1. Click and drag to paint a stroke.

Treadling (red, rows = picks, cols = treadles) — same as tie-up: click to toggle, drag to paint. Multiple treadles per pick are allowed.

The **drawdown** updates live after every edit. The **reconstruction error** display below the drawdown also updates immediately, showing how many pixels differ from the original image. This lets you see whether a manual tweak improves or worsens the approximation.

Picks with no treadle pressed are automatically assigned to the quietest treadle — the one tied to the fewest shafts. This ensures that change-face works correctly on every pick (see Step 6).

Step 6: Change Face

The **Change Face** button inverts every cell of the tie-up, swapping warp-up and weft-up across the entire cloth. The drawdown, reconstruction error, and coloured canvas all update immediately. Clicking it again restores the original face.

In weaving terms, change face switches which colour appears on the face of the cloth. With white warp and blue weft (the default colour scheme), the original face shows white where the warp floats and blue where the weft floats. After change face, the weft is on top wherever the warp was, and vice versa.

For block substitution: The weft colour (blue by default) marks where the pattern weft will appear as thick floats on the face when you do a block substitution such as summer-and-winter. Use change face to choose which regions read as pattern and which as ground before exporting the WIF.

Step 7: Colour schemes and the coloured drawdown

Below the draft, two colour panels let you assign colours to warp threads and weft picks independently. The defaults are all white for warp and all blue for weft.

Preset schemes — click any preset button to apply instantly. Presets use repeating patterns distributed across all threads.

Custom scheme builder — expand the ＋ Create custom scheme link. Choose up to three colours (A, B, C), select a distribution pattern (all A, alternate A/B, stripe of 4, stripe of 3, gradient), preview the result in the strip below, name the scheme, and click ＋ Add to save it as a new preset. The draft must be computed before saving.

Individual thread overrides — click any swatch in the thread list to open a 40-colour palette popup. Use the custom picker at the bottom for any hex colour.

Coloured drawdown — the canvas below the colour panels renders the woven cloth in colour. Warp-up cells show the warp colour; weft-up cells show the weft colour. The canvas updates live as you change any colour. A thin colour strip along the bottom edge shows the warp colour sequence.

Block substitution note — with white warp and blue weft, the blue areas of the coloured drawdown show where you will have pattern if you do a block substitution (e.g. summer-and-winter). The white areas are ground.

Step 8: Downloading the WIF file and notes

Two download buttons appear at the bottom of the page once a draft has been computed.

⬇ **Download WIF file** — saves the draft in Weaving Information File (WIF 1.1) format, compatible with Fiberworks PCW, PixeLoom, and any other WIF-compatible software. The file includes:

- `[WEAVING]` — shaft and treadle count (both equal to Z), rising shed
- `[THREADING]` — one line per warp thread, listing its shaft
- `[TIEUP]` — one line per shaft, listing tied treadles
- `[TREADLING]` — one line per pick, listing pressed treadles
- `[COLOR TABLE]` — all unique colours assigned to warp and weft, as RGB values
- `[WARP COLORS]` and `[WEFT COLORS]` — the colour index for each thread and pick

Orientation: WIF numbers warp threads right-to-left relative to the on-screen display, shafts bottom-to-top, and picks top-to-bottom. These flips are applied automatically so the draft loads correctly in Fiberworks.

⬇ **Download notes** — saves a plain text file containing the image dimensions, Z, full warp and weft colour assignments, an ASCII-art tie-up grid, all six observation answers, and your freeform notes.

Notes tabs — the Notes section (section 8 of the page) has three tabs: Observations (six guided questions about reconstruction error, method comparison, tie-up structure, and colour), Experiments (suggested activities), and My Notes (freeform). Fill these as you explore; they are included in the downloaded notes file.

The mathematics

All five methods solve the same underlying problem: given a binary Y×X image matrix M, find three binary matrices — threading T (Z×X), tie-up U (Z×Z), and treadling R (Y×Z) — such that the Boolean product $T \wedge U \wedge R$ approximates M as closely as possible. The Boolean product is exactly the drawdown: warp x is raised on pick y if any shaft i has T[i][x]=1 and there is a treadle j with R[y][j]=1 and U[i][j]=1.

SVD computes the continuous optimal rank-Z approximation $M \approx U \cdot \Sigma \cdot V^T$ (U is Y×Z, Σ is Z×Z diagonal, V^T is Z×X), then discretises: each column of V^T gives one shaft pattern across the warps (threading), each column of U gives one treadle pattern across the picks (treadling), and Σ gives the tie-up weights. Because SVD is continuous, the binarisation step introduces quantisation error — the gap between the drawdown and the original image.

NMF finds non-negative factors W (Y×Z) and H (Z×X) using multiplicative update rules. Non-negative factors produce additive, parts-based decompositions — each component adds structure rather than cancelling it, which maps naturally to weaving where treadles add raised warps.

BMF works directly on binary matrices, alternating between optimising the shaft pattern (which warps go on each shaft) and the treadle pattern (which picks press each treadle), greedy component by component. It is the closest mathematical match to the physical loom structure.

K-means clusters the X warp column vectors into Z shaft groups and the Y pick row vectors into Z treadle groups independently using Euclidean distance, then derives the tie-up by majority vote over each (shaft, treadle) block.

Alternating optimisation starts from any initialisation and iterates three steps until convergence: (1) fix tie-up and treadling, optimise threading by trying each warp on each shaft and keeping the best; (2) fix tie-up and threading, optimise treadling by trying each pick on each treadle and keeping the best; (3) recompute the optimal tie-up for the current threading and treadling. Each iteration is guaranteed to reduce or maintain reconstruction error.

Comparing methods

The reconstruction error (shown below the drawdown) is the single best guide to which method is working for a given image and Z. Lower is better; 0% means the drawdown exactly reproduces the original image.

Method	Typical reconstruction error	Speed	Notes
SVD	Moderate at low Z; 0% at full rank with Hungarian	Fast	Best understood mathematically; exact at Z = min(X,Y)
NMF	Varies; often better than SVD on sparse images	Moderate	Multiple restarts help considerably
BMF	Often low; depends on image structure	Slower	Strongest for images with clean rectangular blocks
K-means	Moderate; highly dependent on initialisation	Fast	Very sensitive to restarts; run at least 5
Alternating optimisation	Lowest overall, especially at small Z	Slow	Use SVD or K-means init; more restarts = better

Practical workflow:

1. Start with SVD + Hungarian at Z = min(X,Y) to verify 0% error is achievable, confirming the image loaded correctly.
2. Reduce Z to your target shaft count. Compare SVD, K-means, and alternating optimisation at that Z.
3. Enable multiple treadles per pick. This rarely hurts and sometimes substantially reduces error at low Z.
4. Use the manual edit tools to fix any remaining errors that the optimiser missed — a single wrong cell in the tie-up is often responsible for a cluster of errors in the drawdown.
5. Use change face to choose pattern vs ground orientation, then export the WIF.